Confidence intervals: inverting the test

Ciaran Evans

Last time: Formal definition

Let $\boldsymbol{\theta} \in \Theta$ be a parameter of interest, and $X_1, ..., X_n$ a sample from the population. Let $C(X_1, ..., X_n) \subseteq \Theta$ be a set constructed from $X_1, ..., X_n$. Then, $C(X_1, ..., X_n)$ is a $1 - \alpha$ **confidence set** for $\boldsymbol{\theta}$ if

$$\inf_{\boldsymbol{\theta} \in \Theta} P_{\boldsymbol{\theta}}(\boldsymbol{\theta} \in \underbrace{C(X_1, ..., X_n)}_{\text{random set (function of data)}}) = 1 - \alpha$$

So $\forall \theta$, $P_\theta(\theta \in C(X_1, \ldots, X_n)) \geq 1 - \alpha$

Inverting a test

level α test of
$H_0 : \theta = \theta_0$
$H_A : \theta \neq \theta_0$: $P_{\theta_0}((X_1, \dots, X_n) \in \mathcal{R}(\theta_0)) \leq \alpha$

Theorem: Let $\theta \in \Theta$ be a parameter of interest. For each value of $\theta_0 \in \Theta$, consider testing $H_0 : \theta = \theta_0$ vs. $H_A : \theta \neq \theta_0$, and let $\mathcal{R}(\theta_0)$ be the rejection region for a level α test. Let $C(X_1, ..., X_n) = \{\theta_0 \in \Theta : (X_1, ..., X_n) \notin \mathcal{R}(\theta_0)\}$. Then $C(X_1, ..., X_n)$ is a $1 - \alpha$ confidence set for θ.

(construct confidence set by all values θ_0 for which we fail to reject $H_0 : \theta = \theta_0$)

Pf: Suppose $\theta_0 \in C(X_1, \dots, X_n)$

$\theta_0 \in C(X_1, \dots, X_n) \iff (X_1, \dots, X_n) \notin \mathcal{R}(\theta_0)$

$$\Rightarrow P_{\theta_0}(\theta_0 \in C(X_1, \dots, X_n)) = P_{\theta_0}((X_1, \dots, X_n) \notin \mathcal{R}(\theta_0))$$
$$= 1 - P_{\theta_0}((X_1, \dots, X_n) \in \mathcal{R}(\theta_0))$$
$$\geq 1 - \alpha \quad \text{(level } \alpha \text{ test)}$$

So, $\forall\, \theta \in \Theta$, $P_\theta(\theta \in C(X_1, \dots, X_n)) \geq 1 - \alpha$ //

Example

Suppose $X_1, \ldots, X_n \overset{iid}{\sim} Uniform[0, \theta]$. Let's invert the LRT to find a confidence interval for θ.

Consider $H_0: \theta = \theta_0$ vs. $H_A: \theta \neq \theta_0$

LRT: reject when $\Lambda = \dfrac{L(\theta_0 | X)}{L(\hat{\theta} | X)} < K$

$L(\theta | X) = \left(\frac{1}{\theta}\right)^n \mathbb{1}\{X_{(n)} \leq \theta\}$ $\quad \hat{\theta} = X_{(n)}$

$$\Rightarrow \Lambda = \left(\frac{X_{(n)}}{\theta_0}\right)^n \frac{\mathbb{1}\{X_{(n)} \leq \theta_0\}}{\mathbb{1}\{X_{(n)} \leq X_{(n)}\}} = \left(\frac{X_{(n)}}{\theta_0}\right)^n \mathbb{1}\{X_{(n)} \leq \theta_0\}$$

So: reject when $X_{(n)} > \theta_0$ or $\left(\frac{X_{(n)}}{\theta_0}\right)^n < K$

$\Rightarrow$ fail to reject when $X_{(n)} \leq \theta_0$ and

$\left(\frac{X_{(n)}}{\theta_0}\right)^n > K \Rightarrow X_{(n)} > K^{1/n} \cdot \theta_0$

$\Rightarrow \theta_0 < X_{(n)} \cdot c$

fail to reject when $\theta_0 \in [X_{(n)}, c X_{(n)}]$

Example

Suppose $X_1, ..., X_n \overset{iid}{\sim} Uniform[0, \theta]$. Let's invert the LRT to find a confidence interval for θ.

$1-\alpha$ CI for θ: $[X_{(n)}, c X_{(n)}]$

So need c st $P_\theta(X_{(n)} \le \theta \le c X_{(n)}) \ge 1-\alpha$

$$P_\theta(X_{(n)} \le \theta \le c X_{(n)}) = P_\theta(\theta \le c X_{(n)}) = P_\theta\left(\frac{\theta}{c} \le X_{(n)}\right)$$

$$= 1 - P_\theta\left(X_{(n)} \le \frac{\theta}{c}\right)$$

$$= 1 - \left(P_\theta\left(X_i \le \frac{\theta}{c}\right)\right)^n$$

$$= 1 - \left(\frac{\theta/c}{\theta}\right)^n = 1 - \frac{1}{c^n}$$

$$\Rightarrow \quad 1 - \frac{1}{c^n} = 1 - \alpha \quad \Rightarrow \quad c = \frac{1}{\alpha^{1/n}}$$

$$\left[X_{(n)}, \frac{X_{(n)}}{\alpha^{1/n}}\right]$$